

The Riemann Hypothesis

Great Moments in Prime Number Theory: P. L. Chebyshev (1821–1894)

Instead of working directly with the prime counting function

$$\pi(x) = \sum_{p \leq x} 1,$$

Chebyshev introduced a weighted prime counting function

$$\psi(x) = \sum_{p^m \leq x} \log p,$$

where the sum is over prime powers.

For example,

$$\begin{aligned} \psi(30) = & \log 2 + \log 3 + \log 2 + \log 5 + \log 7 + \log 2 + \log 3 + \log 11 + \log 13 \\ & + \log 2 + \log 17 + \log 19 + \log 23 + \log 5 + \log 3 + \log 29. \end{aligned}$$

The function $\psi(x)$ satisfies the fundamental identity

$$\sum_{1 \leq m \leq x} \psi\left(\frac{x}{m}\right) = \sum_{1 \leq n \leq x} \log n,$$

for all positive real numbers x .

Using this identity Chebyshev proved that

$$(0.921)x \leq \psi(x) \leq (1.105)x \quad \text{for all large } x,$$

and from this he obtained the estimate

$$(0.89) \operatorname{li}(x) \leq \pi(x) \leq (1.11) \operatorname{li}(x) \quad \text{for all large } x.$$

Chebyshev also proved that if the limit

$$\lim_{x \rightarrow \infty} \frac{\pi(x)}{\operatorname{li}(x)}$$

exists, then the value of the limit is 1.

A Very Great Moment in Prime Number Theory: B. Riemann (1826–1866)

Riemann wrote one paper of 8 pages on number theory in 1859. In this paper he considered the function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1}.$$

Riemann assumed that

$$s = \sigma + it$$

is a complex variable and observed that the series and product converge to the same analytic function in the half-plane

$$1 < \sigma.$$

In his paper Riemann briefly sketched proofs of his discoveries:

1. The function $\zeta(s)$ has an analytic continuation to the complex plane with a simple pole at $s = 1$.
2. The function $\zeta(s)$ satisfies a functional equation relating its value at s with its value at $1 - s$.
3. The function $\zeta(s)$ has trivial zeros at

$$s = -2, -4, -6, \dots$$

and infinitely many nontrivial zeros in the critical strip

$$\{s = \sigma + it : 0 \leq \sigma \leq 1\}.$$

Riemann also estimated the number of nontrivial zeros:

$$N(T) \sim \frac{T}{2\pi} \log \left(\frac{T}{2\pi} \right) - \frac{T}{2\pi}.$$

Riemann Weighted Prime Counting Function

Riemann defined

$$\Pi(x) = \sum_{p^m \leq x} \frac{1}{m}.$$

He asserted the explicit formula

$$\Pi(x) = \text{li}(x) - \sum_{\rho} \text{li}(x^{\rho}) - \log 2 + \int_x^{\infty} \frac{dt}{(t^2 - 1)t \log t}.$$

A complete proof was given by von Mangoldt in 1895.

A simpler explicit formula is

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log 2\pi - \frac{1}{2} \log(1 - x^{-2}).$$

In both formulas, the summation is over all nontrivial zeros

$$\rho = \beta + i\gamma$$

of $\zeta(s)$.

Equivalent Statements to the Riemann Hypothesis

The following are equivalent to RH:

- (a) All nontrivial zeros satisfy

$$\beta = \frac{1}{2}.$$

- (b) For every $\varepsilon > 0$, there exists $C(\varepsilon) > 0$ such that

$$|\text{li}(x) - \pi(x)| \leq C(\varepsilon)x^{1/2+\varepsilon}.$$

- (c) For every $\varepsilon > 0$, there exists $C(\varepsilon) > 0$ such that

$$|x - \psi(x)| \leq C(\varepsilon)x^{1/2+\varepsilon}.$$

- (d) Define the Möbius function $\mu(n)$ by

$$\mu(n) = \begin{cases} (-1)^m, & \text{if } n \text{ is a product of } m \text{ distinct primes,} \\ 0, & \text{if } n \text{ is divisible by the square of a prime.} \end{cases}$$

Then

$$\lim_{N \rightarrow \infty} \sum_{n=1}^N \frac{\mu(n)}{n^{\sigma}}$$

converges for all $\sigma > \frac{1}{2}$.

Prime Number Theorem

In 1896 Hadamard and de la Vallée Poussin proved that

$$\zeta(1 + it) \neq 0 \quad \text{for all real } t.$$

From this they proved the Prime Number Theorem:

$$\lim_{x \rightarrow \infty} \frac{\pi(x)}{\text{li}(x)} = 1.$$

Littlewood's Theorem

In 1914 J. E. Littlewood proved that

$$\text{li}(x) - \pi(x)$$

changes sign infinitely often.

Thus this difference is both positive and negative for arbitrarily large x .

Selberg's Result

In 1942 Selberg proved that a positive proportion of nontrivial zeros lie on the critical line.

If $N(T)$ is the number of zeros

$$\rho = \beta + i\gamma, \quad 0 < \gamma \leq T,$$

and $N_0(T)$ counts zeros with real part exactly $1/2$, then

$$cN(T) \leq N_0(T) \quad \text{for all large } T,$$

for some positive constant c .

Selberg and Erdős

In 1948 Selberg and Erdős found elementary proofs of the Prime Number Theorem.

Their proof does not use:

- the Riemann zeta function,

- complex analysis,
- Fourier analysis.

Montgomery's Pair Correlation

In 1970 Montgomery showed that if RH is true, then the imaginary parts of the nontrivial zeros have the same pair correlation statistics as eigenvalues of random Hermitian matrices.